

Yixiang Wang

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Google Scholar | GitHub | LinkedIn

Education

Carnegie Mellon University , Pittsburgh, PA	Aug 2024 – Present
M.S. in Electrical and Computer Engineering Current GPA: 4.0/4.0	
Rensselaer Polytechnic Institute , Troy, NY	Aug 2020 – May 2024
B.S. in Electrical Engineering GPA: 3.99/4.0 Dean's Honor List Magna Cum Laude	
The Chinese University of Hong Kong Shenzhen , Shenzhen, Guangdong	Fed 2023 - Apr 2023
Visiting Student at the Robotic Institute	

Research Interests

I have broad interests in AI and machine learning, with a primary focus on **online learning** and **online optimization** and their applications in domains such as **fraud detection** and the integration of **quantum computing**. My research aims to develop robust learning systems that can effectively handle **concept drift**, **domain adaptation**, and **data imbalance**. In parallel, I am exploring **continual learning** in deep models, particularly language models, to understand how they can adapt to evolving data streams. Currently, I am investigating how **small language models** can achieve continuous learning through **reinforcement learning**, enabling them to outperform larger models on specialized tasks such as task decomposition and action planning.

Publications

[1] J. Wang, **Y. Wang**, et al., *Transformable Inspection Robot Design and Implementation for Complex Pipeline Environment*, IEEE Robotics and Automation Letters, June 2024.

Research Experience

Online Fraud Detection CMU, May 2025 – Present
Advisor: Prof. Ozan Tonguz

- Surveyed fraud-detection methods, with emphasis on online learning and graph neural network approaches.
- Currently developing a robust online fraud-detection system capable of handling concept drift, class imbalance, and real-time decision making.
- Exploring the integration of quantum machine learning in fraud detection as a promising direction, examining high-dimensional representations as well as other quantum advantages beyond classical methods.

Potential of Language Model-Powered Multi-Agent Collaboration CMU, Feb 2025 – Present
Advisor: Prof. Ozan Tonguz

- Built a multi-agent collaboration platform with an LLM-driven control framework.
- Compared centralized and decentralized coordination and assessed long-term memory effects on performance.
- Improved task success rate by 18% through prompt and context engineering.
- Currently developing methods for small language models to surpass larger ones on targeted tasks via reinforcement learning, RAG, and knowledge-graph integration while reducing token and compute costs.
- Currently exploring a language-model-based decision-action architecture for task decomposition, planning, and inter-agent communication to improve generalization to complex scenarios.

Design and Implementation of Transformable Robots for Pipeline Inspection [1] CUHKSZ, Feb - Aug 2023
Advisor: Dr. Yongquan Chen

- Designed a transformable robot with pneumatic end-effectors and CV system for deformation and obstacle detection.
- Built a real-time control system linking the air pump with the robot to enable automated actuation.
- Achieved a 191.4% maximum deformation rate, allowing multiple stable postures and high adaptability.
- Conducted extensive experiments in complex terrains to validate locomotion and deformation performance.
- Published results in *IEEE Robotics and Automation Letters*, 2024.

Industry Experience

Software Engineering Intern, ASML, San Jose, CA Jun 2024 – Aug 2024

- Designed Ticket-Guided CI/CD workflow, enabling automated Bamboo builds parameterized by Jira ticket data

- Built a Ticket-Guided-Build system with Python, Shell, and REST APIs to automate software builds
- Tested the TGB system for the DevOps team through 20 trials before deploying to production
- Improved development workflow efficiency and empowered non-technical users to trigger builds
- Directed FOSS planning, establishing workflows and ensuring open-source license compliance
- Presented a FOSS implementation analysis to Brion Team, outlining timelines, phases, and stakeholders

Teaching Experience

Teaching Assistant, RPI, *Electrical Circuits (ECSE 2010)* Fall 2022

- Conducted weekly lab sessions and office hours for 60+ students.
- Graded assignments and guided students on circuit design projects and provided constructive feedback.
- Assisted professor in improving course materials and refining syllabus.

Teaching Assistant, CMU, *Applied Stochastic Processes (ECE 18751)* Fall 2025

- Led weekly recitations to reinforce lecture concepts and guide students in solving practical problems.
- Held weekly office hours and provided detailed feedback while grading assignments.
- Reported class performance to the professor after each exam and recommended strategies for improvement.
- Assisted with course logistics, including course materials management and attendance tracking.

Awards

- Dean's Honor List, RPI, 2020–2024

Technical Skills

- **Programming:** Python, Pytorch, C/C++, Shell, Hugging Face
- **ML/AI Frameworks:** Transformer, LLM, CNN, LSTM
- **DevOps Tools:** Docker, Bamboo, Jira, Git, Linux